

Informe LAB O2 Haskell

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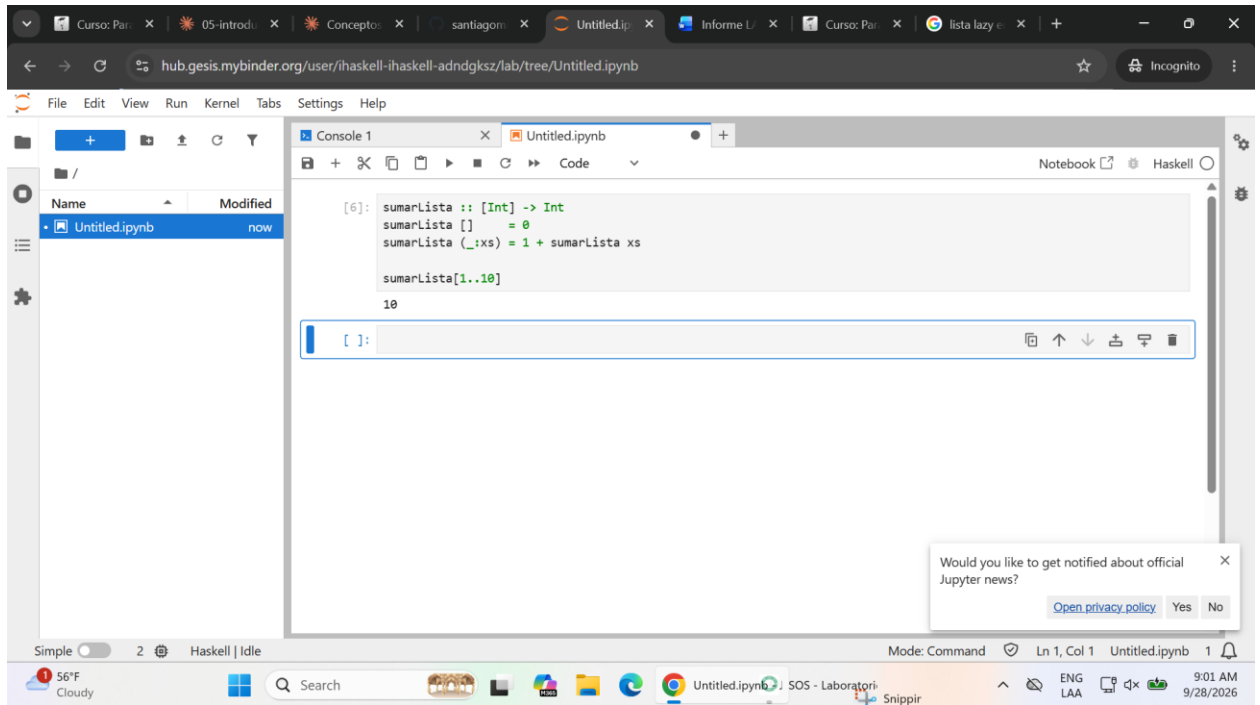
Paradigmas de la Programación

Juan Camilo Ángel Hernández

de de 2026

Ejercicio 1 - Manejo de listas

- 1) Definir las siguientes funciones usando recursión.
 - a) Retorna la longitud de una lista



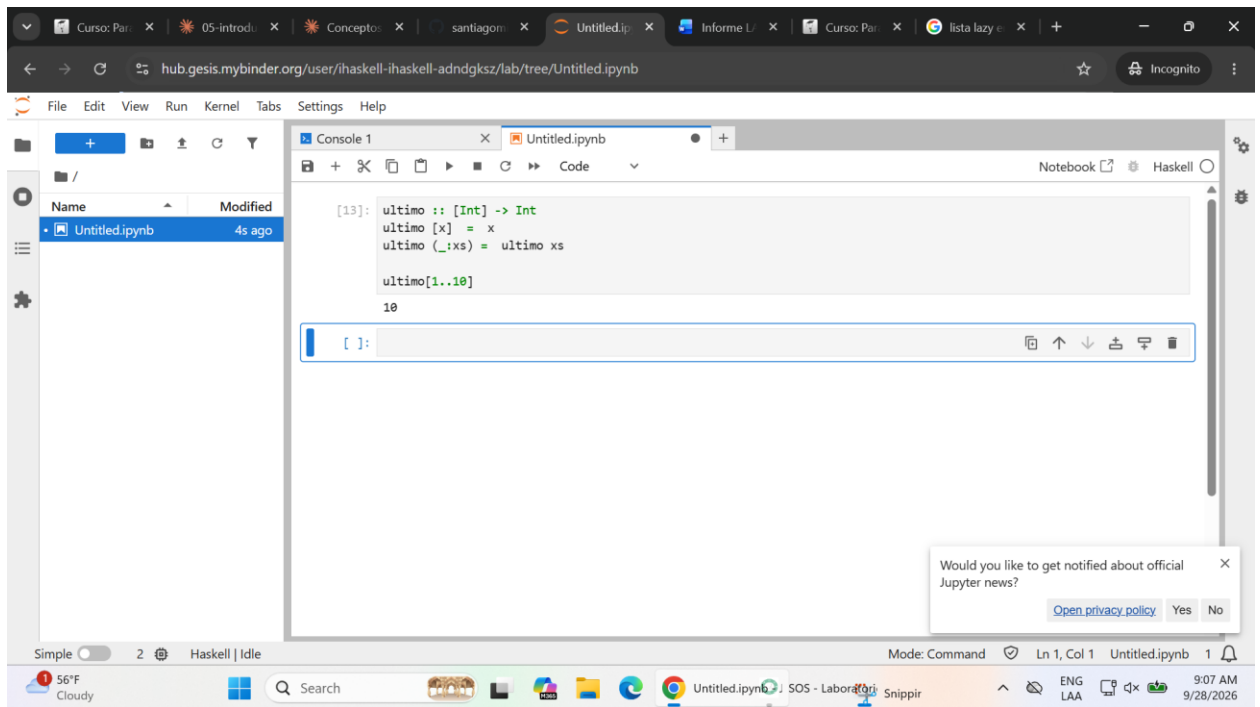
The screenshot shows a web browser window with a Jupyter Notebook interface. The browser tabs include 'Curso: Par...', '05-introdu...', 'Concepto...', 'santiagom...', 'Untitled.ip...', 'Informe L...', 'Curso: Par...', and 'lista lazy...'. The address bar shows the URL 'hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgks/lab/tree/Untitled.ipynb'. The notebook interface has a menu bar with 'File', 'Edit', 'View', 'Run', 'Kernel', 'Tabs', 'Settings', and 'Help'. On the left, a file explorer shows 'Untitled.ipynb' as the active file, modified 'now'. The main editor area shows the following Haskell code:

```
[6]: sumarLista :: [Int] -> Int
    sumarLista [] = 0
    sumarLista (_:xs) = 1 + sumarLista xs

    sumarLista [1..10]
```

The output of the code is '10'. A small notification box in the bottom right corner asks 'Would you like to get notified about official Jupyter news?' with links for 'Open privacy policy', 'Yes', and 'No'. The status bar at the bottom indicates 'Simple', '2', 'Haskell | Idle', 'Mode: Command', 'Ln 1, Col 1', 'Untitled.ipynb', and the time '9:01 AM 9/28/2026'.

- b) Retorna el último elemento de una lista



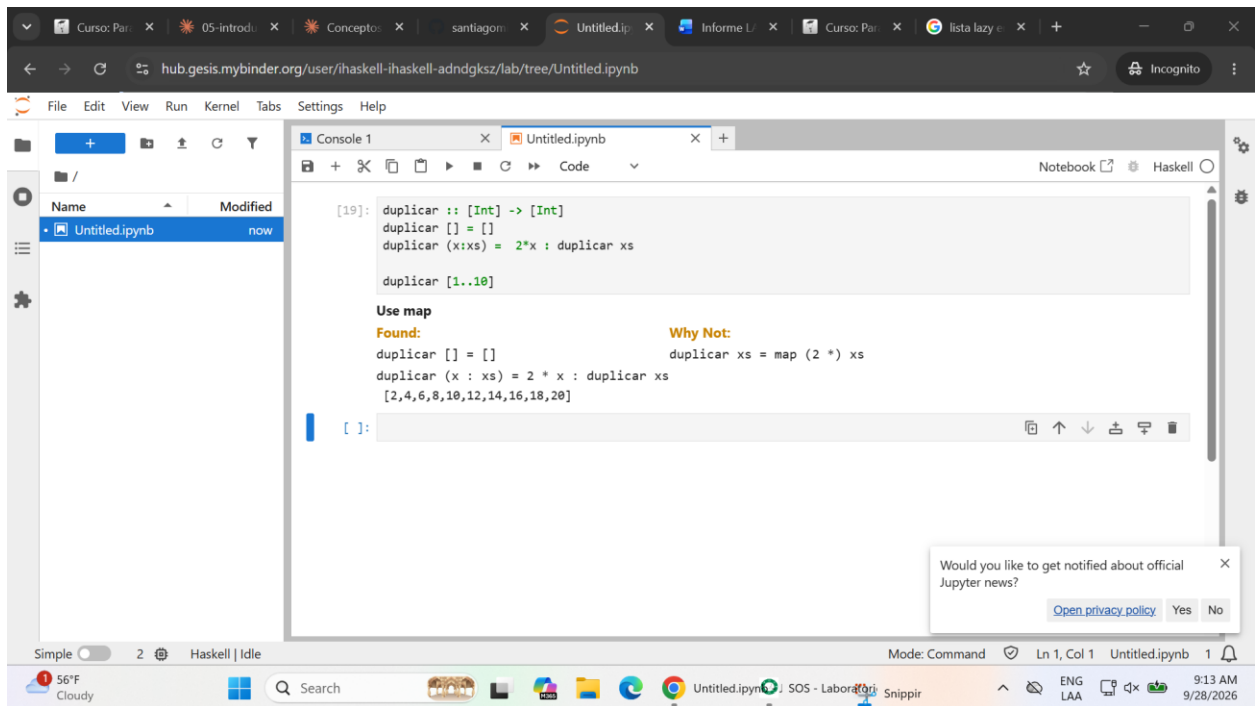
The screenshot shows a web browser window with a Jupyter Notebook interface, similar to the one above. The browser tabs and address bar are the same. The notebook interface shows the following Haskell code:

```
[13]: ultimo :: [Int] -> Int
      ultimo [x] = x
      ultimo (_:xs) = ultimo xs

      ultimo [1..10]
```

The output of the code is '10'. The same notification box and status bar are present at the bottom. The status bar indicates 'Simple', '2', 'Haskell | Idle', 'Mode: Command', 'Ln 1, Col 1', 'Untitled.ipynb', and the time '9:07 AM 9/28/2026'.

c) Duplica cada elemento de una lista de números



The screenshot shows a Jupyter Notebook interface in a web browser. The browser's address bar displays the URL: `hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb`. The notebook's left sidebar shows a file explorer with a single file named `Untitled.ipynb`. The main area of the notebook contains the following Haskell code:

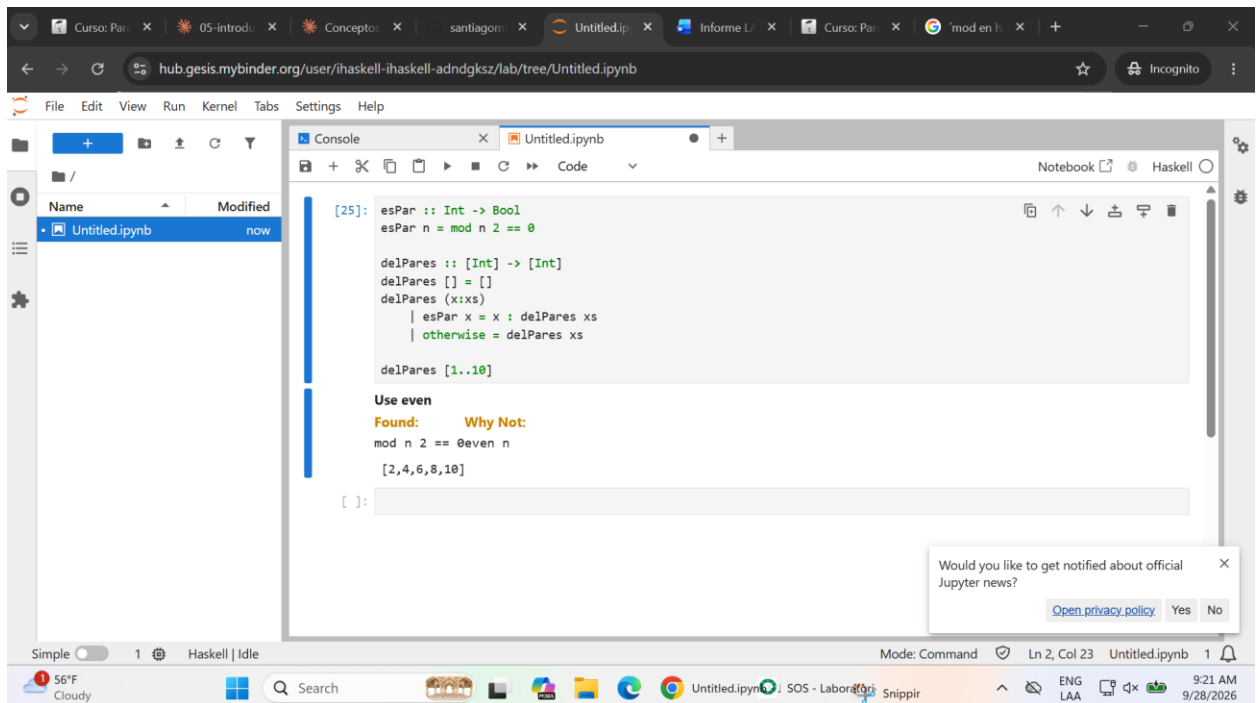
```
[19]: duplicar :: [Int] -> [Int]
      duplicar [] = []
      duplicar (x:xs) = 2*x : duplicar xs

      duplicar [1..10]

      Use map
      Found:          Why Not:
      duplicar [] = []      duplicar xs = map (2 *) xs
      duplicar (x : xs) = 2 * x : duplicar xs
      [2,4,6,8,10,12,14,16,18,20]
```

Below the code, there is a prompt: `[ ]:`. At the bottom of the notebook, a status bar indicates the mode is `Command`, the current line and column are `Ln 1, Col 1`, and the file is `Untitled.ipynb`. A small notification box in the bottom right corner asks: "Would you like to get notified about official Jupyter news?" with links for `Open privacy policy`, `Yes`, and `No`.

d) Eliminar los elementos pares de una lista



The screenshot shows a Jupyter Notebook interface in a web browser. The browser's address bar displays the URL: `hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb`. The notebook's left sidebar shows a file explorer with a single file named `Untitled.ipynb`. The main area of the notebook contains the following Haskell code:

```
[25]: esPar :: Int -> Bool
      esPar n = mod n 2 == 0

      delPares :: [Int] -> [Int]
      delPares [] = []
      delPares (x:xs) =
        | esPar x = x : delPares xs
        | otherwise = delPares xs

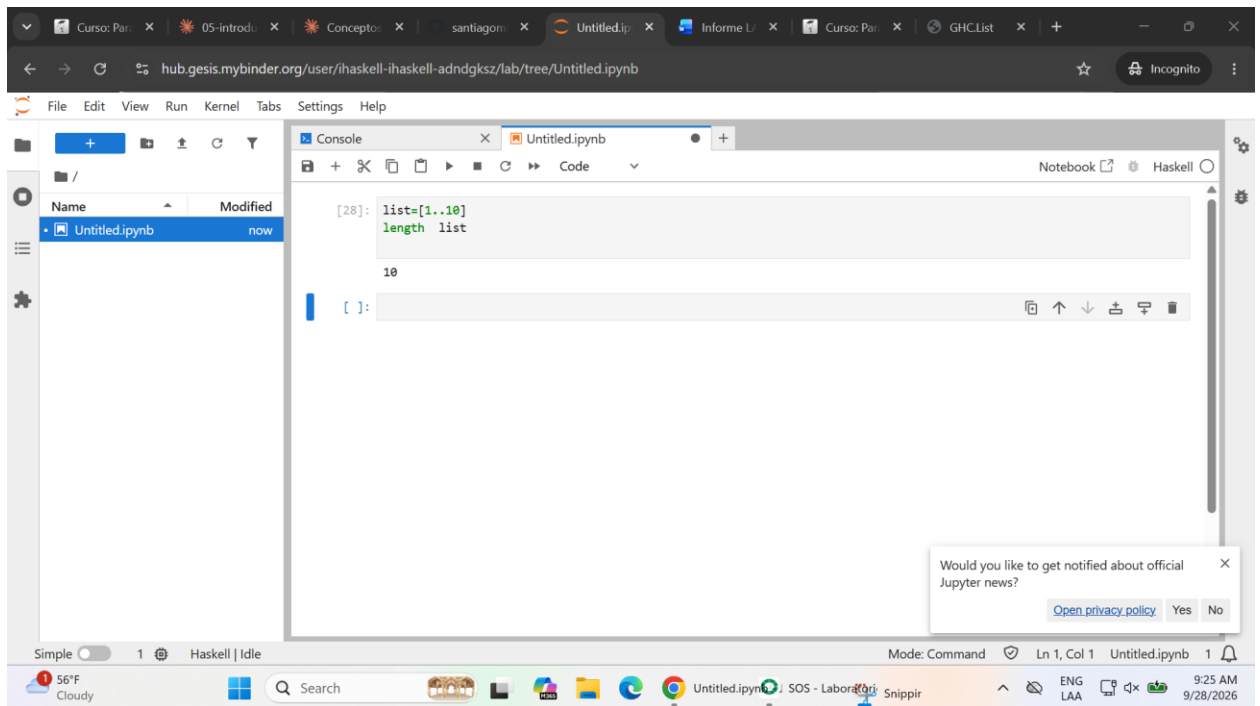
      delPares [1..10]

      Use even
      Found:          Why Not:
      mod n 2 == 0 even n
      [2,4,6,8,10]
```

Below the code, there is a prompt: `[ ]:`. At the bottom of the notebook, a status bar indicates the mode is `Command`, the current line and column are `Ln 2, Col 23`, and the file is `Untitled.ipynb`. A small notification box in the bottom right corner asks: "Would you like to get notified about official Jupyter news?" with links for `Open privacy policy`, `Yes`, and `No`.

2) Define estas mismas funciones pero ahora usando funciones de orden superior como `map` y `filter`

a) Retorna la longitud de una lista



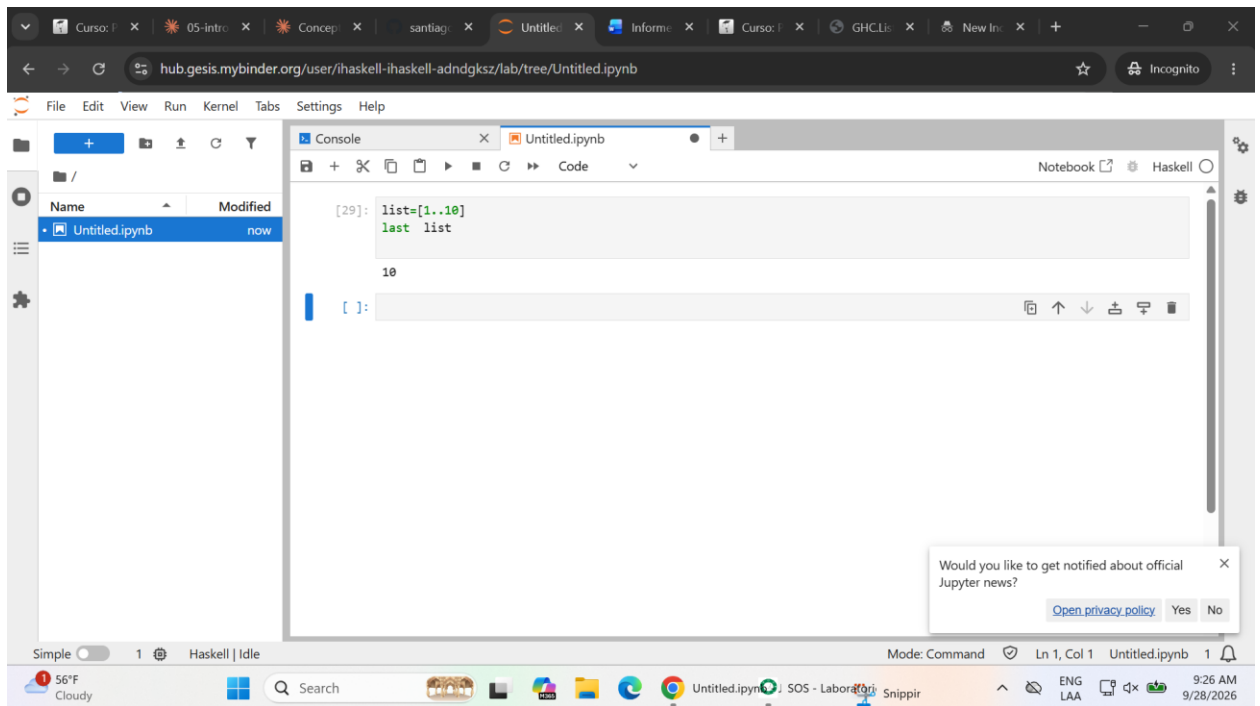
The screenshot shows a Jupyter Notebook interface with a file browser on the left and a code editor on the right. The code editor contains the following code:

```
[28]: list=[1..10]
      length list

10
```

The output of the code is 10. A notification dialog box is visible in the bottom right corner, asking "Would you like to get notified about official Jupyter news?". The browser address bar shows the URL: hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb.

b) Retorna el último elemento de una lista



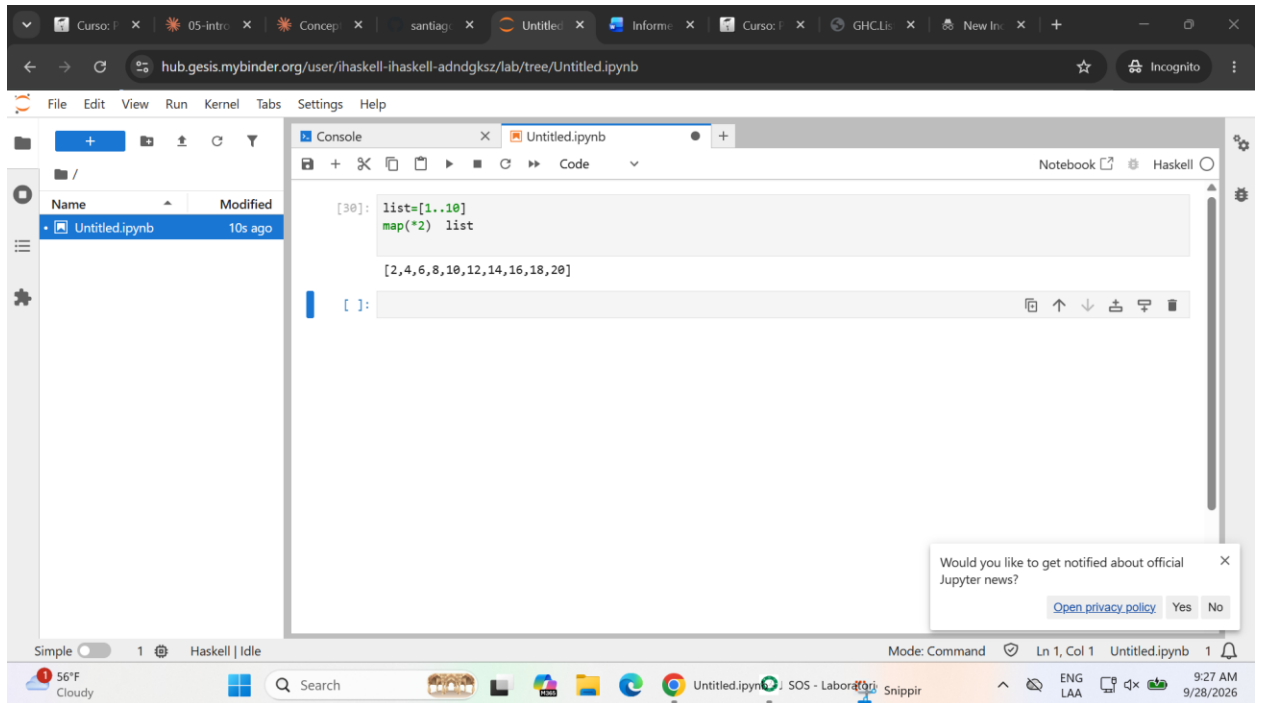
The screenshot shows a Jupyter Notebook interface with a file browser on the left and a code editor on the right. The code editor contains the following code:

```
[29]: list=[1..10]
      last list

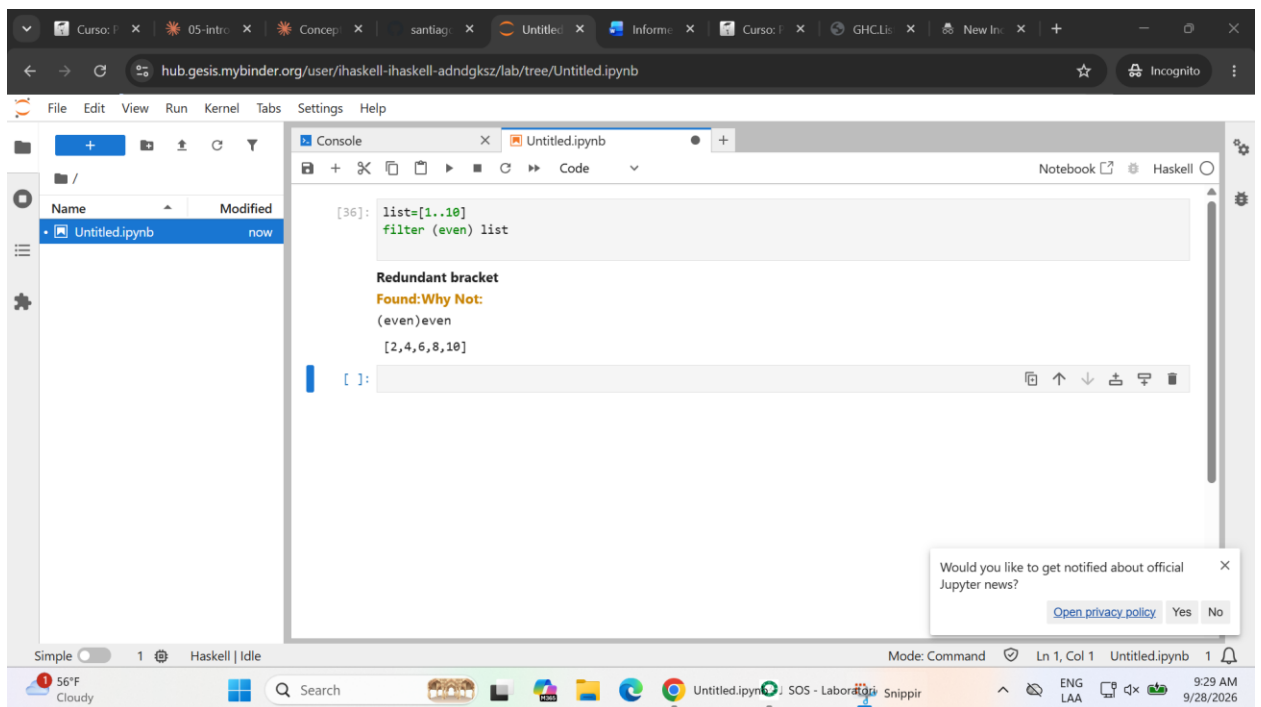
10
```

The output of the code is 10. A notification dialog box is visible in the bottom right corner, asking "Would you like to get notified about official Jupyter news?". The browser address bar shows the URL: hub.gesis.mybinder.org/user/ihaskell-ihaskell-adndgksz/lab/tree/Untitled.ipynb.

c) Duplica cada elemento de una lista de números



d) Eliminar los elementos pares de una lista



Ejercicio 2 - Listas por comprensión

1) Usa comprensión de listas para:

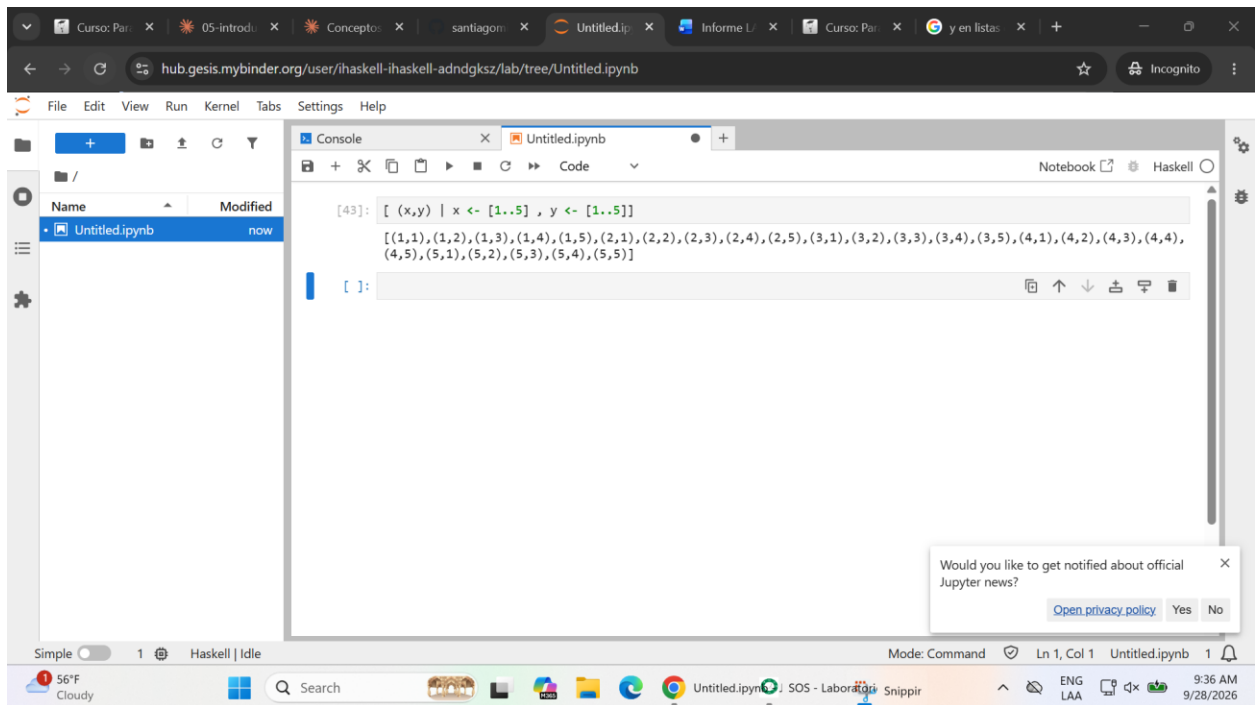
a) Obtener los cuadrados de los primeros 10 números.

The screenshot shows a Jupyter Notebook in a web browser. The left sidebar displays a file explorer with 'Untitled.ipynb' listed. The main area contains a code cell with the following Haskell code: `[39]: [ x^2 | x <- [1..10] ]`. Below the code, the output is displayed as `[1,4,9,16,25,36,49,64,81,100]`. A notification bubble in the bottom right corner asks, 'Would you like to get notified about official Jupyter news?' with 'Yes' and 'No' buttons. The bottom status bar shows 'Haskell | Idle' and the system clock as 9:31 AM on 9/28/2026.

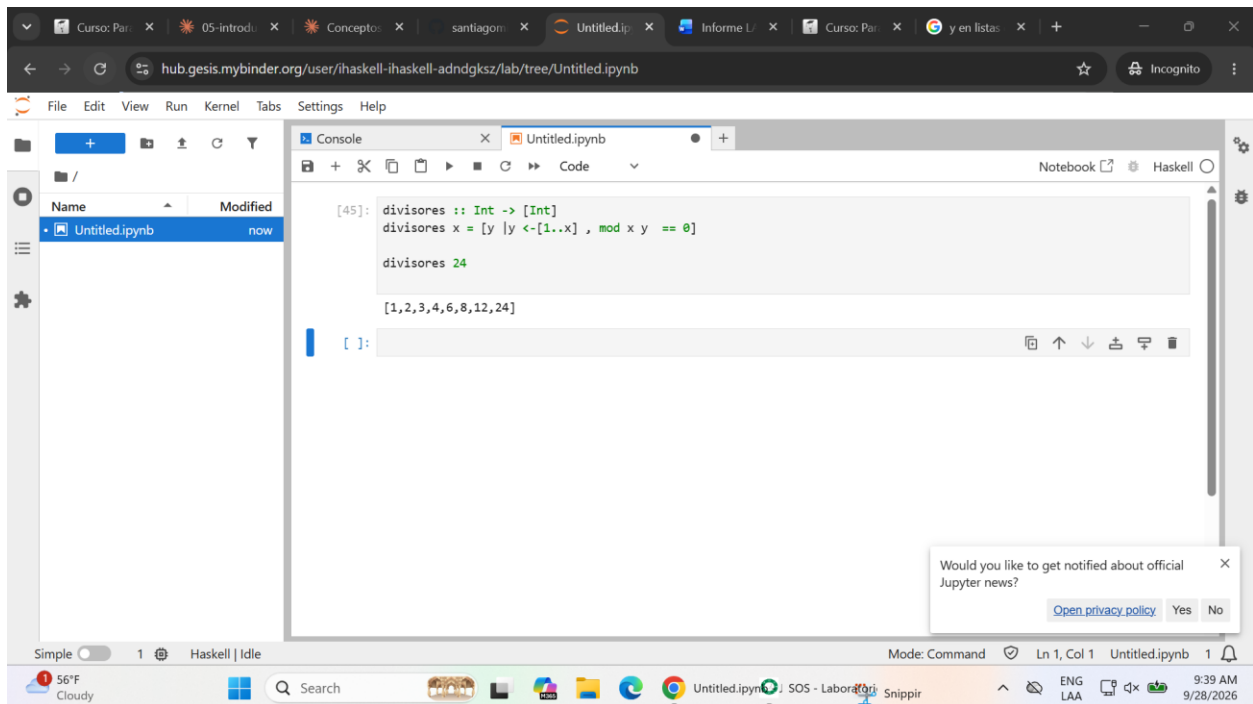
b) Filtrar los divisibles por 3 entre 1 y 30.

The screenshot shows the same Jupyter Notebook interface. The code cell now contains: `[42]: [ x | x <- [1..30] , mod x 3 == 0 ]`. The output is `[3,6,9,12,15,18,21,24,27,30]`. The notification bubble and status bar are identical to the previous screenshot.

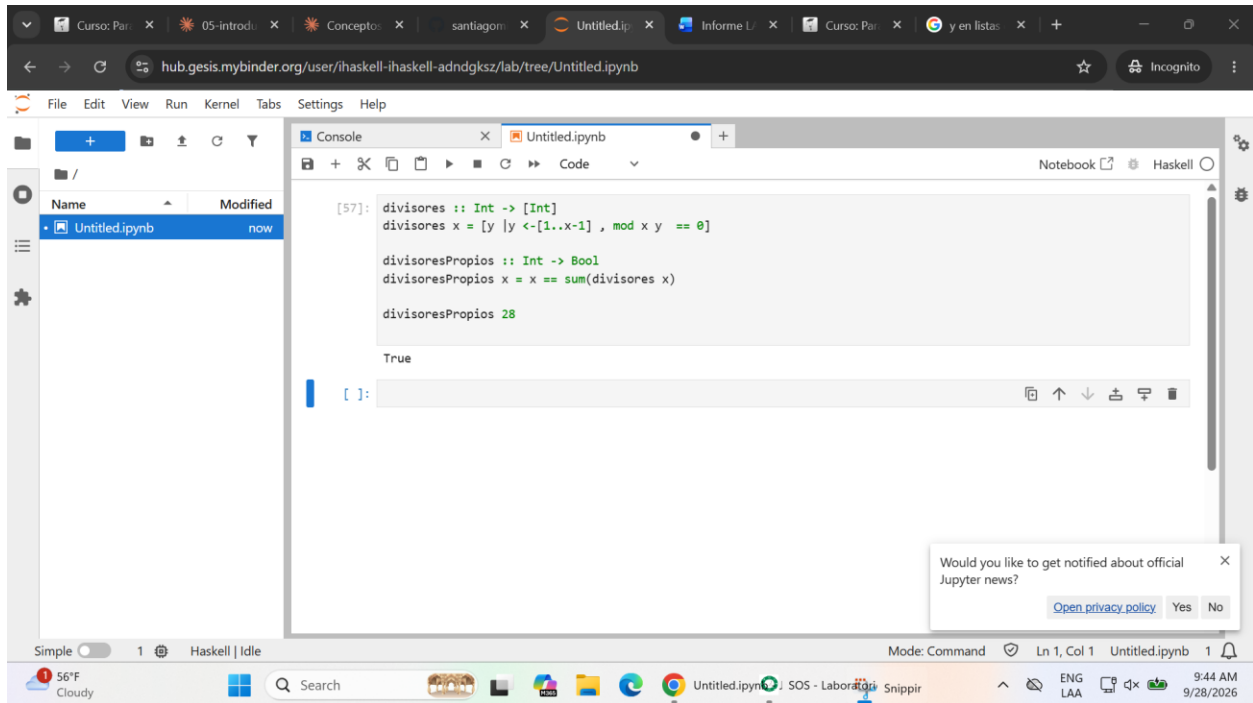
c) Generar una lista de pares (x, y) donde $x < y$ y ambos estén en `[1..5]`.



- 2) Define una función que devuelva los divisores de un número (usando lista por comprensión):

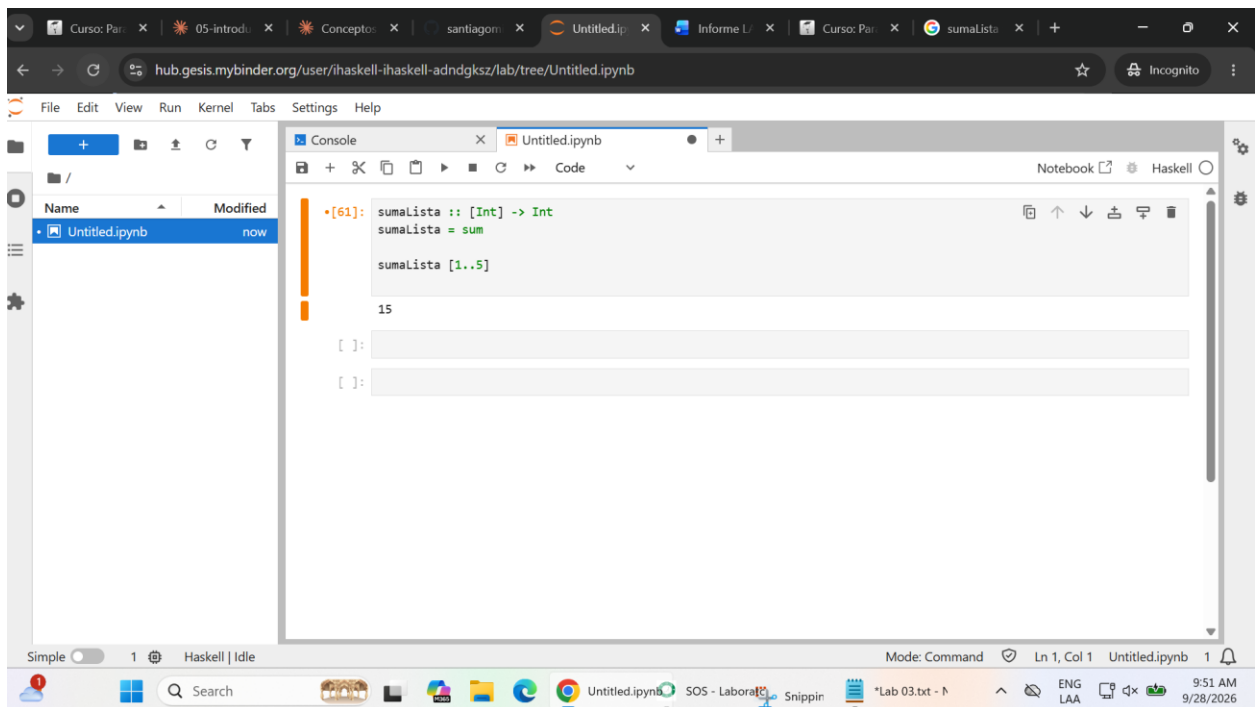


- 3) Define la función divisoresPropios, crea una función que determine si un número es perfecto (suma de sus divisores propios = número):

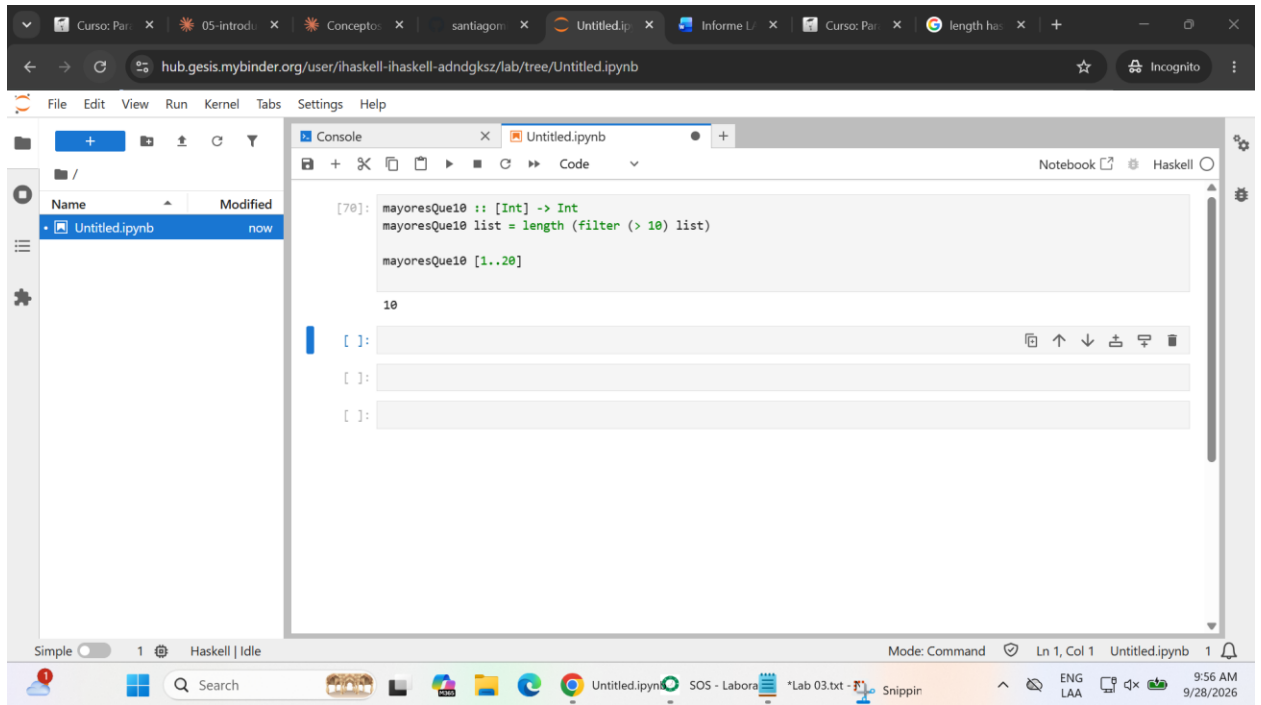


Ejercicio 3 - Funciones de orden superior

- 1) Reescribe las siguientes funciones sin recursión explícita, usando funciones de orden superior:
 - a) Suma de todos los elementos



- b) Contar cuántos elementos son mayores que 10



c) a

2) Define las siguientes funciones usando filter